or poor visual displays may overwhelm operators, reducing reaction speed and situational awareness. As autonomy increases, maintaining human-in-the-loop (HITL) control becomes even more challenging, especially when multiple drones are managed simultaneously.

To mitigate these issues, researchers are developing adaptive remote control systems that employ AI-driven decision support tools to assist operators. These systems can automatically prioritize alerts, suggest corrective actions, or temporarily assume control during emergencies. Improvements in 5G connectivity, low-latency satellite communication, and redundant control channels are also enhancing reliability. Future control systems will likely emphasize shared autonomy, where humans and drones dynamically adjust their levels of control based on situational demands, ensuring both safety and efficiency.

7.10.3 Enhancing user experience in drone operation for various industries

A positive user experience (UX) is fundamental for widespread adoption of drones across industries such as logistics, agriculture, filmmaking, public safety, and infrastructure inspection. The effectiveness of drone systems depends not only on performance and autonomy but also on how intuitively and comfortably humans can operate them. Poorly designed interfaces or overly technical controls can limit the accessibility of drones, particularly for non-expert users.

Enhancing UX begins with simplifying operational workflows. Automated flight planning, one-touch takeoff and landing, and intelligent mission templates reduce user effort and the potential for human error. For industrial operators, customized control dashboards tailored to specific applications, such as crop monitoring or thermal inspection, can streamline operations and improve productivity.